

- **Preliminary Results**

This analysis was born from real frustration. After getting overcharged at an auto repair shop with no explanation of what was done or why, we wanted to know, is this just bad luck, or is this happening to everyone? We collected 11,022 Yelp reviews from 32 Bay Area auto repair shops and used data science to find out. The answer is clear: it's not just you.

- **H1: Pricing Transparency is a Primary Driver of Negative Sentiment**

1. Summary

When we looked at what people actually complain about in 1 and 2 star reviews, pricing came up the most. Using topic modeling on 656 negative reviews, we found that the largest single complaint category was what we called "Pricing & Labor", 189 reviews out of 656 fell into this bucket. These are reviews where customers talk about being quoted one price and charged another, feeling like they were pushed into repairs they didn't need, or simply having no idea whether the price they paid was fair. This topic also had the most negative emotional tone of all four categories, with an average VADER sentiment score of -0.242 . In plain terms, the reviews about pricing aren't just negative, they're the angriest ones in the dataset.

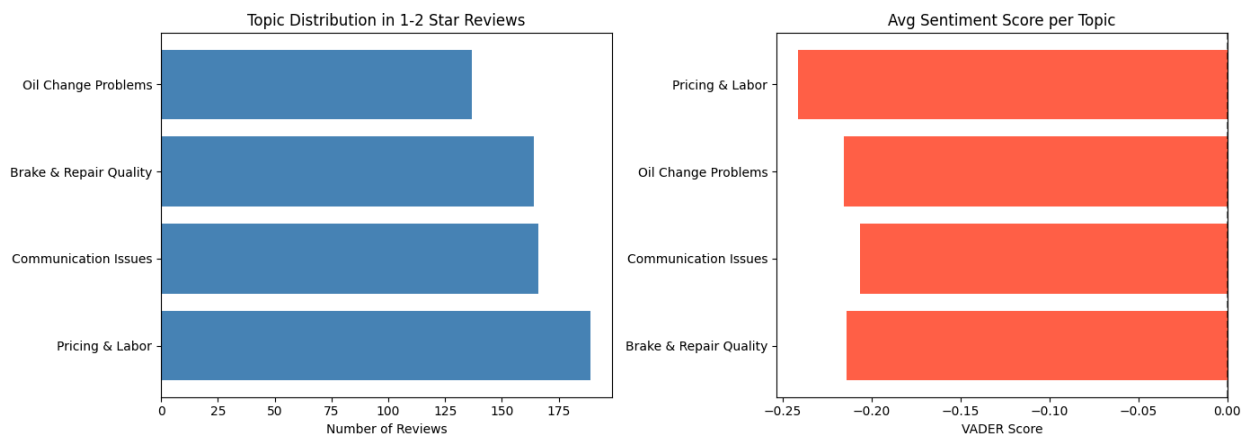
2. Implications

This confirms the hypothesis. Pricing is not just one of many complaints, it is the loudest and most emotionally charged one. What makes it especially damaging is that customers often don't even know they're being overcharged at the moment. They only realize it later, or they have a gut feeling that something wasn't right. That gap between what a shop charges and what a customer can verify is exactly the problem this project set out to expose. The data shows it is real, it is

common, and it affects thousands of people across the Bay Area every year. From a broader perspective, this hits hardest for people who don't have the automotive knowledge to push back, or who can't afford to get a second opinion. The information asymmetry between mechanics and customers is not a small inconvenience, it's a systemic issue that the data clearly reflects.

3. Next Steps:

The next step is to go deeper, extracting actual price mentions from reviews to quantify how often quoted prices differ from final charges. We also want to build a per-shop scoring system that flags businesses with the highest concentration of pricing complaints.



- **H2: Communication Quality Strongly Influences Customer Satisfaction**

1. Summary

The second most common complaint wasn't about a bad repair. It was about being left in the dark. Our topic modeling identified "Communication Issues" as the second largest category with 166 reviews and an average sentiment of -0.206 . The words that define this topic tell the whole story: "call", "called", "never", "know", "issue". Customers who never got a callback. Customers who weren't told what was wrong with their car. Customers who had to keep following up just to get a

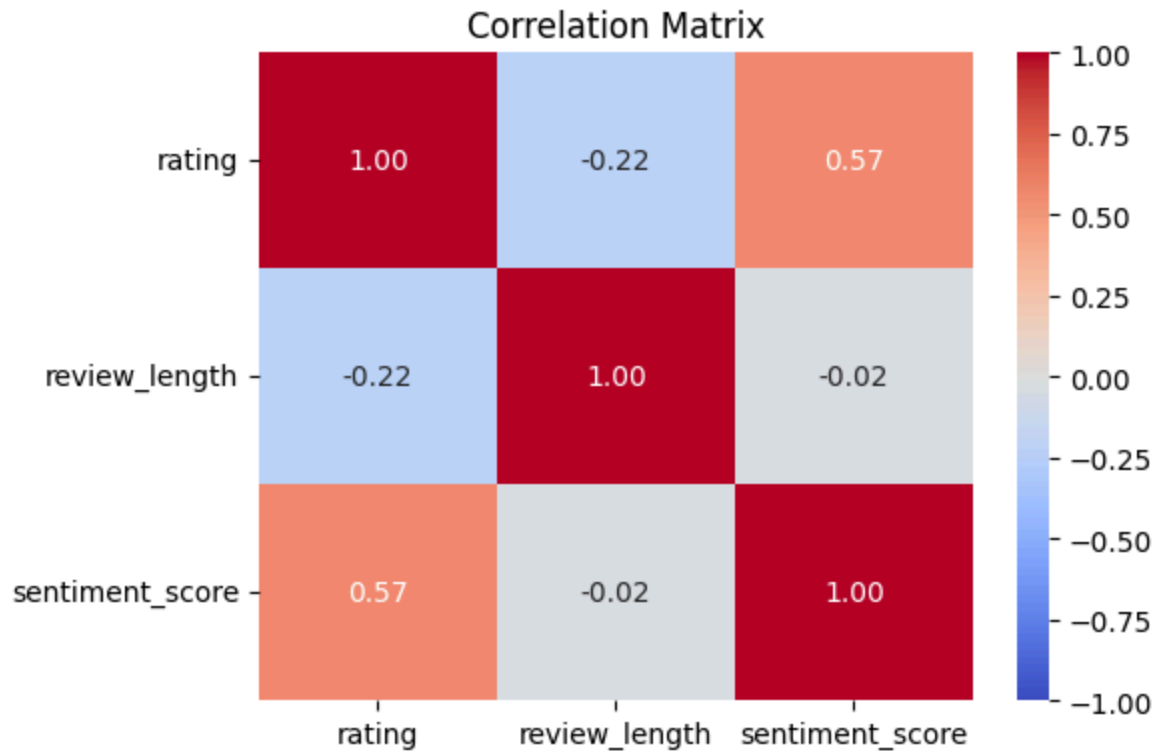
status update. To further validate this, we ran a correlation analysis between star ratings and VADER sentiment scores across all 11,022 reviews. The result was $r = 0.57$... a strong positive relationship confirming that the emotional tone of a review reliably matches how many stars a customer gave. The more negative the language, the lower the rating. This held true consistently across all shops.

2. Implications

What this tells us is that a bad repair is not the only thing that makes someone leave a 1-star review. Being ignored, being confused, or being treated like your questions don't matter is enough on its own. Customers who feel powerless, who don't understand what's wrong with their car and aren't helped to understand, express that powerlessness in their reviews, and they express it loudly. This is important because it means fixing the communication experience alone could meaningfully improve how customers feel about a shop, even if the mechanical work doesn't change. Transparency in how a shop communicates, clear estimates, regular updates, plain-language explanations of the problem matters just as much as the quality of the repair.

3. Next Steps

We want to isolate communication-specific reviews and test whether shops that actively respond to negative reviews on Yelp show improved ratings over time. This could show whether engagement itself builds trust.



- **H3: A Small Number of Complaint Categories Dominate Negative Feedback**

1. Summary

If you read enough 1-star auto repair reviews, you start to notice the same stories over and over. Our analysis confirmed that this isn't just an impression, it's a measurable pattern. Topic modeling on 656 negative reviews produced exactly 4 coherent complaint categories: Pricing & Labor (189 reviews), Communication Issues (166), Brake & Repair Quality (164), and Oil Change Problems (137). Every single one of these topics carries a negative average sentiment score. There are no neutral topics, every recurring theme in negative reviews represents real anger and real frustration. What's striking is how evenly distributed these four topics are. No single issue crushes the others. Instead, the industry

is failing customers in four consistent, predictable ways, all at roughly the same frequency.

2. Implications

This is the finding that ties everything together. The auto repair industry doesn't have a few bad actors with unique problems. It has a small set of systemic failures that show up across dozens of shops, thousands of reviews, and multiple cities. That consistency is actually good news from a solutions perspective, because if the problems are predictable, so are the solutions. Societally, this matters because these aren't just annoying experiences. For someone who depends on their car to get to work, an unexpected \$800 bill or a repair that didn't fix the problem can be genuinely harmful. The more we can arm drivers with information before they walk into a shop, the more we level a playing field that has been tilted against them for a long time.

